Jafar Banar

Senior Data & AI Engineer | Python · LLM · ML · 5G/6G | PhD candidate, Telecom & Radar (Chalmers)

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Swedish permanent residence and EU long-term resident status (unrestricted right to work in Sweden)

PROFESSIONAL SUMMARY

Data and AI engineer with 10+ years in telecom and radar R&D and in industry data and AI roles. Most recently a consultant at Volvo Cars on a production test-data pipeline; before that built an internal on-prem LLM service at QRTech and fixed a radar perception bug at Einride. PhD candidate at Chalmers and TU Eindhoven (defence February 2027) in synchronization for distributed MIMO and radar.

CORE SKILLS

- **Daily use:** Python, SQL, LLM/RAG, embeddings and vector search, PyTorch, scikit-learn, XGBoost, FastAPI, Docker, Git, GitHub Actions, Linux, MATLAB, signal processing
- **Working knowledge:** LoRA/QLoRA fine-tuning (A100, SLURM), AI agents and MCP, LangGraph, ONNX/OpenVINO, Azure/AKS, Kubernetes, PostgreSQL, Qdrant, FAISS/ChromaDB, Grafana, C/C++, Go, ROS1
- **Domains:** Automotive radar and LiDAR, sensor fusion, CAN (interface specification), ISO 26262/SOTIF (awareness), 4G/5G, 3GPP, OFDM, distributed MIMO, SDR/USRP
- **Languages & Licence:** English (Fluent), Swedish (A2, SFI C completed, studying SFI D) · Driving licence B

PROFESSIONAL EXPERIENCE

Senior Data Engineer (Consultant) May 2026 – Present
Sigma Embedded Engineering Yora AB, *Gothenburg, Sweden*

- Client assignment at **Volvo Cars** (May to Sep 2026) on the production pipeline that synchronises measurement data from engine, e-motor, HIL and battery test rigs.
- Made the pairing of two measurement systems deterministic, so each test gets one reproducible pair instead of failing silently; validated on historical runs and merged.
- Replaced hard deletes with a reversible, quarantine-first cleanup, added a timeout to hanging report runs, and onboarded a new dSpace HIL rig in QA with no regression.
- Since Sep 2026, building AI demos for Sigma, including a retrieval-based support assistant and an OpenVINO vision demo.

Tech: *Python, SQL, ASAM MDF4, Sympathy for Data, dSpace HIL, Azure/AKS, GitHub Actions, Grafana*

Senior Development Engineer (AI Engineer) Feb 2025 – Apr 2026
QRTech AB, *Gothenburg, Sweden*

- Built an internal on-prem LLM service on a CPU-only server (four 4-bit open models on llama.cpp, FastAPI gateway, Open WebUI, FAISS retrieval), opened to all staff with 5–10 regular users, and tuned it, cutting short-prompt response time from 5–8 s to about 3 s.
- Contributed to a team-built LangGraph pipeline (called multi-agent in the code) that generates test specifications from standards, building its standards-upload API, document classification, documentation and demo.
- Built document anonymization (regex, spaCy and Microsoft Presidio, incl. Swedish personal identity numbers) and an AI CV-matching workflow in n8n.
- Wrote the requirements, interface and verification documents for an agricultural robot MVP and contributed to its hazard analysis (HARA).
- Client assignment at **Einride** as Perception Engineer (consultant), Sep to Dec 2025: found and fixed a Go perception bug that under-reported radar target speed by $\cos^2$(elevation) (3.0 m/s read as 0.75 m/s at 60°); validated on 22 recorded scenarios and merged into production code.
- Einride: analysed radar velocity errors and LiDAR timing against OXTS ground truth, and contributed to a SOTIF LiDAR+radar feasibility prototype.

Tech: *Python, FastAPI, llama.cpp, LangGraph, FAISS/ChromaDB, Qdrant, PostgreSQL, spaCy, Presidio, n8n, Docker, Kubernetes; Einride: Go, C++, Python, ROS1, LCM, PyTorch, OXTS*

Founder (part-time) Jul 2024 – Present
AIOpt AB, *Gothenburg, Sweden*

- Building a safety gate for AI-proposed changes to mobile networks: rule-based checks that approve, refuse or send each change to a person, with signed, hash-chained records and an open-source verifier (github.com/aiopt-se/aiopt-verify).
- Lab-validated on public O-RAN data and an srsRAN/Open5GS 5G software stack, including a tool-using operations agent whose changes must pass the gate; seeking a first operator pilot. Almi verification funding.

Visiting Researcher, Automotive Radar (PhD secondment) Jun 2021 – Nov 2022

NXP Semiconductors, *Eindhoven, Netherlands*

- Synchronization and signal processing for coherent distributed automotive radar (ADAS research), seconded from Chalmers as a Marie Skłodowska-Curie fellow (EU MyWave).
- Developed a maximum-likelihood clock-drift estimator for two bistatic 78 GHz FMCW radars, presented at IEEE ISPCS 2024.

Tech: *MATLAB, C++, Python, LabVIEW, signal processing, automotive radar*

Doctoral Researcher & Project Assistant (Wireless Systems) Jul 2020 – Jan 2025
Chalmers University of Technology, *Gothenburg, Sweden*

- Derived closed-form spectral efficiency of distributed massive MIMO under phase noise (EuMC 2022) and developed a user-centric phase synchronization method (+85% 5th-percentile spectral efficiency in simulation; journal paper in preparation).
- Built MATLAB/Python simulators and ran USRP experiments; presented at EuMC 2022 and IEEE ISPCS 2024, and gave an invited talk at IEEE ISIT 2025.
- Co-supervised 15 thesis students (7 MSc and 2 bachelor projects) with partners including Saab, Ericsson and Qamcom; TPC member, IEEE WCNC 2024–2026 and IEEE ISAC 2026.

Tech: *MATLAB, Python, SDR/USRP, distributed MIMO, OFDM, synchronization, calibration, beamforming*

RF Optimization Engineer Nov 2019 – Jun 2020
AEC - RavanErtebat (Access Endless Communication)

- Optimized a regional cluster of a live Ericsson 2G/3G/4G network for a large mobile operator: KPI reporting, parameter trials, NodeB audits and trouble-ticket analysis.

Tech: *Ericsson RAN (RNC/NodeB), TEMS Discovery, Linux, Bash, KPI analysis, drive testing*

Telecom Researcher Apr 2016 – Nov 2019
ICT Research Institute

- Researched and simulated 5G and wireless technologies (massive MIMO, full duplex, satellite links, IoT) in MATLAB, Python and C++; wrote technical reports and gave internal trainings.

Tech: *MATLAB, Python, C++, 3GPP/ETSI standards, simulation*

SELECTED PROJECTS

- CaseTrace (public, github.com/JafarBanar/casetrace): a read-only forensics AI agent with typed tools over MCP and a bounded plan, verify and self-correct loop that blocks findings without supporting evidence.
- groundcite (PyPI): a Python library that checks an LLM answer cites only its given sources and fails CI otherwise; extracted from a 3GPP NR retrieval assistant with BM25 written from scratch.
- LoRA fine-tuning of Qwen2.5-7B on an A100 (Chalmers Alvis) for telecom root-cause classification: 66.7% vs 40% for rules; a simpler XGBoost model reached 74.9% and was recommended.
- Mentor-AI (ENHANCE European university alliance): AI mentor matching with TF-IDF, assignment optimization and a local LLM check, piloted on real mentor–mentee pairs; co-awarded a Diversity Seed Fund grant.

EDUCATION

PhD, Electrical Engineering (Dual Degree) Jul 2020 – Feb 2027 (expected)
Chalmers University of Technology / TU Eindhoven, *Sweden / Netherlands*

- Thesis: array calibration and synchronization for distributed massive MIMO and radars, written alongside full-time industry work since 2025; defence February 2027.
- MSc, Electrical Engineering, IUST, 2014–2016: full-duplex cooperative OFDMA networks (journal paper 2017).
- BSc, Electrical Engineering, AUT, 2009–2014: data transmission protocols for LEO satellite links.
- 3rd place, IEEE ISIT 2025 machine-learning challenge (CSI-based localization).